

BMEG 3105 Fall 2026

Data analytics for personalized genomics and precision medicine

Scribing lecture 4: From sequence to gene expression matrix: Assembly and mapping

1. How to do Dynamic programming

- Need a scoring matrix.
- Fill in the DP table, preserving arrows.
- Last cell = best alignment score.
- Trace back arrows to get the alignment.

1. Gene Expression Matrix

- Human genomes are very similar:
 - Genetic variation accounts for only ~0.001% of the genome.
 - Only ~1% of the human genome encodes protein.
- Gene expression differences may explain phenotype differences.
- Workflow to obtain a gene expression matrix:
 1. mRNA isolation
 2. Illumina sequencing
 3. Align sequences against the genome
 4. Count reads per gene
 5. → Gene expression matrix

2. Genome Assembly

- Illumina sequencing read length: ~200 bp.
- Human genome length: $\sim 3 \times 10^9$ bp.
- Genome is assembled from overlapping short reads ~ solving a jigsaw puzzle.

- Two short reads can overlap. The genome is reconstructed from these overlaps.
- Example:
 - Reads: TAA, AAT, ATG, TGC, GCC, CCA, CAT, ATG, TGG, GGA, GAT, ATG
 - ➔ TAATGCCATGGATG
- Challenges in assembly:
 - Mutations
 - Conflicts (e.g., AAT vs AAA)
 - Repeat sequences (e.g., AAAAAAAAAA...)
 - Repeat genes (e.g., AATAATAATAAT) and copy number determination
 - Algorithm speed

3. Sequence Mapping

- Goal: map short reads back to a reference genome.
- Example:
 - Genome: TAATGCCATGGATG
 - Reads: TAA, CCA, GAT, GCC, CCA, ATG
- Method: slide each read along the genome and calculate the difference.
- Dynamic programming can be used to calculate differences

4. More About Dynamic Programming

- Global alignment: only cares about the value in the last cell; aligns entire sequences.
- Local alignment: finds similar components, motifs, and domains within dissimilar sequences.
- Scoring matrix:
 - Mismatch models mutations.
 - Gap models insertion/deletion and gene duplications.
- Many scoring matrices exist (e.g., BLOSUM), built from different databases; choice depends on the problem.

- Gap penalty depends on how sequence similarity is defined.
 - Example: if similarity is defined as “the number of all matched pairs,” the scoring matrix and gap penalty change accordingly.
- DP table stores sub-problem answers and construction path (arrows), but for global alignment only the last cell’s value is needed.